Performance Evaluation

What's performance?

$$\text{Energy} = (\text{capacitive load}) \times (\text{Voltage})^2$$

Power - rate of consumption of energy

$$= \frac{\text{Energy}}{\text{time}}$$

$$= \frac{(\text{capacitive load}) \times (\text{voltage})^2}{\text{time}}$$

$$= (\text{capacitive load}) \times (\text{Voltage})^2 \times \text{freq}$$

1985 microproc became dominant

transistors get faster as they get smaller

Dennard scaling rule

Moore said transistors get cheaper 2x per every 24 mo

"More's law" - computers doubled in speed ~ every 18 months

1995 - wires connecting transistors became dominant

↳ pipelining, instruction-level parallelism

2005 Power ~ V^2 x freq

1 volt



Pentium 4 Prescott

Two notions of time performance

DC to Paris via jet

Plane	DC to Paris	Passangers	speed	P. Mph
Boeing 747	6.5 hours	470	610mph	286,700
Concorde	3 hours	132	1350 mph	178,200

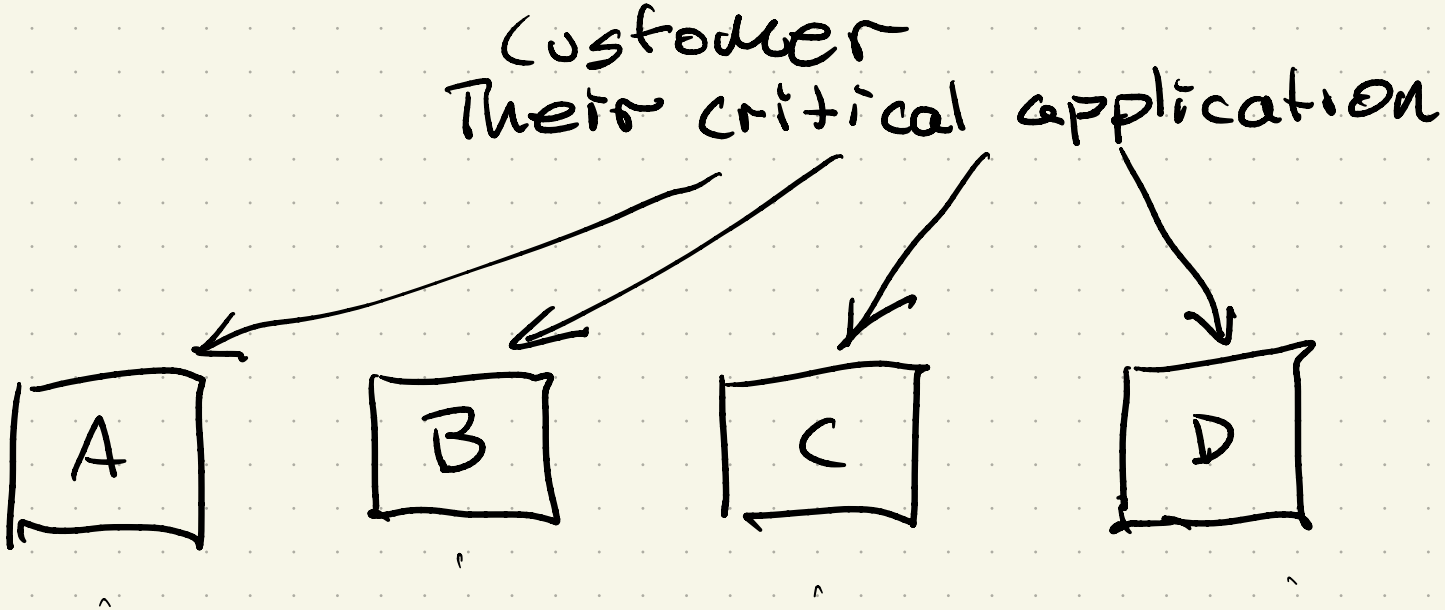
latency
Turnaround time

Throughput
Bandwidth

How do you measure performance of a proposed computer architecture?

1. Build it
   - costly, slow
2. Model performance
   - Mathematical - queuing systems, Markov processes, Petri nets ...
   - Simulation
     1. Circuit level SPICE
     2. Gate level (0, 1, X, Z)
     3. Functional - "timer"
        - Given a set of instructions, determine total # of clock cycles needed to execute them

What workloads to use for simulation?



Benchmarks

SPEC cpu SPEC89, 92, 95, 2000
2006, 2017
"engineering workstation"

EEMBC
Geekbench
High-Performance LINPACK (HPL) Supercomputer 500
" Conjugate Gradient (HPCG)

Transaction Processing Performance Council
TPC-A, TPC-C, ...
No code in these benchmarks!

Time performance

CPU runtime - single program
remove context switching costs

CPU time = total clock cycle count x Cycle time

$$= \frac{\text{clock cycle count}}{\text{frequency}}$$

clock cycle count = instruction count (I.C.)
x average cycles per instruction

$$\text{CPU time} = \text{I.C.} \times \overline{\text{CPI}} \times \text{C.T.}$$